

A Proposal For Mach-Zehnder Interferometer

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I. INTRODUCTION

In recent years, breakthroughs in AI have fueled explosive growth in compute demand. Ever more compute resources are required to train larger LLM models. To train such modes in reasonable amount of time, faster GPUs and larger GPU clusters along with the high-speed data communication networks that cover up to a hundred meters are required. At this communication distance and speed, traditional copper cables are no longer feasible, and high speed optical links are required. Silicon photonic eletro-optical modulators are the key component that enables high-speed optical data communication networks. In particular, Mack-Zender Interferometer (MZI) based silicon photonics modulator is widely used for its high-speed capability and low cost. In this report, theory and modelling of basic MZI design is discussed. Several MZI prototypes without heater and electrically controlled phase shifter are fabricated using ebeam technology. The measurement result and comparison between measurement and simulation is discussed.

II. THEORY

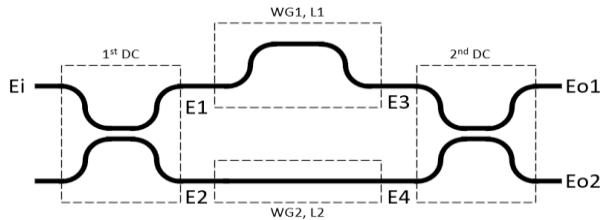


Fig 1: Schematic of MZI core

In this prototype design, we choose 1550nm TE mode MZI's with 3 difference

branch length differences (ΔL). The core of MZI consists of 2 broadband 3dB directional couplers (DC), 2 wave guides of different lengths (WG1 and WG2), and 2 grating couplers to bring light in and out of the chip. As shown in Fig 1, the first DC splits the input light (E_i) into 2 guided beams and feeds them into 2 imbalanced waveguides (E1, E2). The second DC combines the 2 light beams coming out of the 2 waveguides (E3, E4). The input and output lights are coupled off-chip using 3 1550nm TE mode grating couplers, one for the input, and 1 for MZI's output (E_{o1}), and 1 for MZI's complementary output (E_{o2}). The output of the 1st DC can be related to the input light using the following equation [1]:

$$\begin{bmatrix} E_1 \\ E_2 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -j \\ -j & 1 \end{bmatrix} \cdot \begin{bmatrix} E_i \\ 0 \end{bmatrix} \quad (1)$$

After the 2 light beams passing through WG1 and WG2, their outputs experience a phase shift according to the WG length (assuming lossless WG):

$$\begin{bmatrix} E_3 \\ E_4 \end{bmatrix} = \begin{bmatrix} e^{-j\frac{2\pi n_e L_1}{\lambda}} & 0 \\ 0 & e^{-j\frac{2\pi n_e L_2}{\lambda}} \end{bmatrix} \cdot \begin{bmatrix} E_1 \\ E_2 \end{bmatrix} \quad (2)$$

The outputs of the MZI can be related to E_3 , E_4 as the following:

$$\begin{bmatrix} E_{o1} \\ E_{o2} \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -j \\ -j & 1 \end{bmatrix} \cdot \begin{bmatrix} E_3 \\ E_4 \end{bmatrix} \quad (3)$$

Combine equations (1), (2), and (3):

$$\begin{bmatrix} E_{o1} \\ E_{o2} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & -j \\ -j & 1 \end{bmatrix} \begin{bmatrix} e^{-j\frac{2\pi n_e L_1}{\lambda}} & 0 \\ 0 & e^{-j\frac{2\pi n_e L_2}{\lambda}} \end{bmatrix} \begin{bmatrix} 1 & -j \\ -j & 1 \end{bmatrix} \begin{bmatrix} E_i \\ 0 \end{bmatrix}$$

The transfer functions from input to both outputs are:

$$\begin{bmatrix} E_{o1} \\ E_{o2} \end{bmatrix} = -j \begin{bmatrix} \sin\left[\frac{\pi n_e}{\lambda}(L_1 - L_2)\right] \\ \cos\left[\frac{\pi n_e}{\lambda}(L_1 - L_2)\right] \end{bmatrix} e^{-j\frac{\pi n_e}{\lambda}(L_1 + L_2)} E_i$$

Define $\Delta L = L_1 - L_2$, the light intensity of outputs are:

$$\begin{bmatrix} I_{o1} \\ I_{o2} \end{bmatrix} = \begin{bmatrix} \sin^2\left(\frac{\pi n_e \Delta L}{\lambda}\right) \\ \cos^2\left(\frac{\pi n_e \Delta L}{\lambda}\right) \end{bmatrix} \cdot I_i$$

For both outputs, free spectral range, $\Delta\lambda_{FSR}$, must satisfy the following equation:

$$\frac{\pi n_e \Delta L}{\lambda} - \frac{\pi(n_e + \frac{dn_e}{d\lambda} \Delta\lambda_{FSR}) \Delta L}{\lambda + \Delta\lambda_{FSR}} = \pi \quad (4)$$

Solving Equation (4):

$$\left(n_e - \frac{dn_e}{d\lambda} \lambda\right) \Delta\lambda_{FSR} = \frac{\lambda^2}{\Delta L} \left(1 + \frac{\Delta\lambda_{FSR}}{\lambda}\right)$$

Assuming $\Delta\lambda_{FSR} \ll \lambda$,

$$\left(n_e - \frac{dn_e}{d\lambda} \lambda\right) \Delta\lambda_{FSR} \approx \frac{\lambda^2}{\Delta L}$$

Using the definition of the group index $n_g = n_e - \frac{dn_e}{d\lambda} \lambda$ [2], we have:

$$\Delta\lambda_{FSR} \approx \frac{\lambda^2}{\Delta L n_g} \quad (5)$$

Equation (5) is used to calculate expected FSR for the design. In this design, ΔL is chosen to be 114.53 μm , 76.355 μm , and 57.266 μm for three different MZI circuits. Using $\lambda = 1.55\mu\text{m}$ and $n_g = 4.1953$ obtained by finite-difference eigen mode simulation, the expected FSR can be summarized in Table I below:

Table I: ΔL vs. FSR for designed MZI's

	ΔL	FSR
MZI1	114.53 μm	5nm
MZI2	76.355 μm	7.5nm
MZI3	57.266 μm	10nm

III. MODELING AND SIMULATION

In this design, 500nm wide and 220nm wide slab waveguide surrounded by SiO_2 cladding is used. We chose the 1st TE mode and the primary mode of design. The mode profile for the 1st TE mode is simulated by Lumerical finite-difference eigen mode solver is plotted in Fig 2. The simulated effective index and group index are plotted in Fig 3.

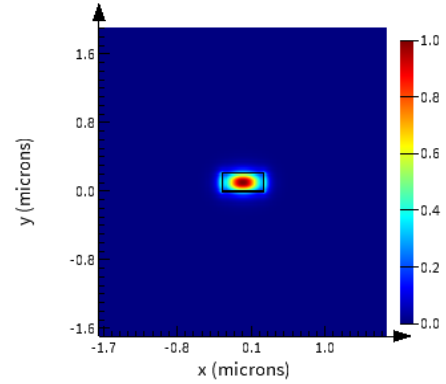


Fig 2: Simulated mode profile for W/H=500nm/220nm 1550nm TE mode waveguide

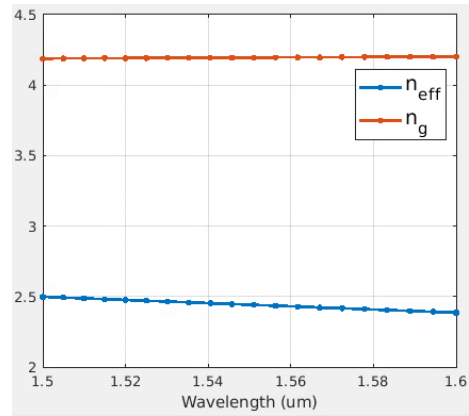


Fig 3: Simulated effective index (real part) and group index versus wavelength

The effective index and group index for the 1st TE mode of this waveguide is plotted as a function of wavelength in Fig 4a and 4b. The simulation effective index versus wavelength is fitted to a 2nd order polynomial function to obtained waveguide compact model:

$$n_e = 2.4419 - 1.1312(\lambda - 1.55\mu\text{m}) - 0.0422533(\lambda - 1.55\mu\text{m})^2$$

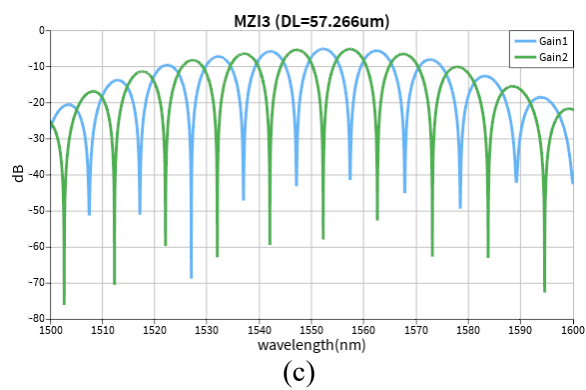
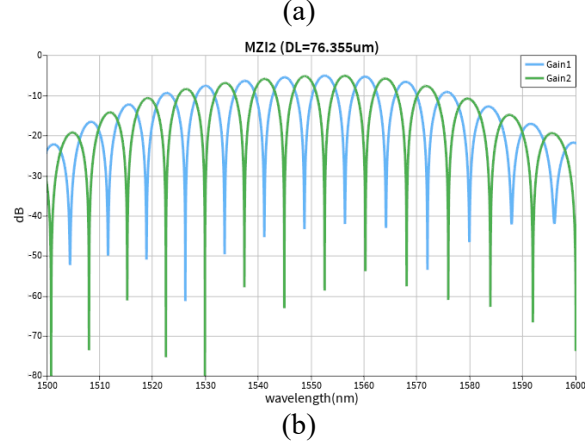
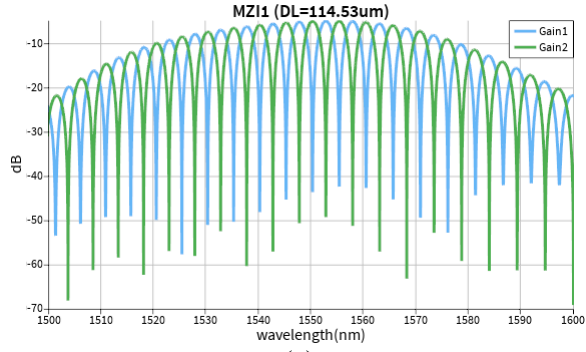


Fig. 4: Simulated Spectrum for (a) MZI1 with $\Delta L=114.53\mu\text{m}$, (b) MZI2 with $\Delta L=76.335\mu\text{m}$, and (c) MZI3 with $\Delta L=57.226\mu\text{m}$

Fig. 4 shows the simulated gain of MZI's. The FSR calculated from these plots are summarized in Tabel II below.

Table II: Designed vs. Simulated FSR of MZI's

	ΔL	FSR	
		Designed	Simulated
MZI1	114.53 μm	5nm	5.05nm
MZI2	76.355 μm	7.5nm	7.62nm
MZI3	57.266 μm	10nm	10.19nm

REFERENCE

- [1] NPTEL-NOC IITM. (2022, October 11). Integrated Optical Components: DC based MZI and Microring Resonator (MMR) [Video]. YouTube. <https://www.youtube.com/watch?v=V3C5zCNY97Y&t=17s>
- [2] Chrostowski and Hochberg. Silicon Photonics Design (Cambridge University Press, 2015), 66.